Appendix A. Search Strategies for electronic databases

MEDLINE (Ovid) search strategy

1. deglutition disorders/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,kf.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,kf.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,kf.
5. (feeding* or regurgit*).tw,kf.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,kf.
7. or/1-6
8. oropharynx/ or oropharyn*.tw,kf.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,kf.
10. 8 and 9
11. 7 or 10
12. electric stimulation therapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,kf.
14. 12 or 13
15. exp child/ or exp "congenital, hereditary, and neonatal diseases and abnormalities"/ or exp infant/ or adolescent/ or exp pediatrics/ or child, abandoned/ or exp child, exceptional/ or child, orphaned/ or child, unwanted/ or minors/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

Embase (Ovid) search strategy

1. dysphagia/ or swallowing/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,kw.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,kw.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,kw.
5. (feeding* or regurgit*).tw,kw.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,kw.
7. or/1-6
8. oropharynx/ or oropharyn*.tw,kw.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,kw.
10. 8 and 9
11. 7 or 10
12. electrotherapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,kw.

14. 12 or 13
15. juvenile/ or exp adolescent/ or exp child/ or exp postnatal development/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

PsycINFO (Ovid) search strategy

1. dysphagia/ or swallowing/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw,id.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw,id.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw,id.
5. (feeding* or regurgit*).tw,id.
6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw,id.
7. or/1-6
8. oropharyn*.tw,id.
9. (dysphag* or disorder* or dysfunc* or impair*).tw,id.
10. 8 and 9
11. 7 or 10
12. electrical stimulation/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw,id.
14. 12 or 13
15. adolescent development/ or childhood development/ or pediatrics/ or exp Congenital Disorders/ or child characteristics/ or child abuse/ or exp child welfare/ or chronically ill children/ or child neglect/ or child psychiatry/ or child psychopathology/ or exp child care/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre term or preterm* or premature birth or NICU or preschool* or pre school* or kindergarten* or elementary school* or nursery school* or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or prepubesc* or pre pubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

Cochrane Central Register of Controlled Trials (OVID) search strategy

1. deglutition disorders/
2. (dysphagia* or deglutition or swallow or swallows or swallowing or odynophagia*).tw.
3. (deglut* adj3 (abnormal* or disorder* or dysfunc* or impair*)).tw.
4. ((oropharynx* or trachea* or lung* or pulmon*) adj3 aspirat*).tw.
5. (feeding* or regurgit*).tw.

6. (pharyng* adj3 (dysphag* or dysfunct* or disorder* or impair*)).tw.
7. or/1-6
8. oropharynx/ or oropharyn*.tw.
9. (dysphag* or disorder* or dysfunc* or impair*).tw.
10. 8 and 9
11. 7 or 10
12. electric stimulation therapy/
13. (electrostimulation or NMES or electrical stimulation or electric stimulation or electrotherapy or electro therapy).tw.
14. 12 or 13
15. exp child/ or exp "congenital, hereditary, and neonatal diseases and abnormalities"/ or exp infant/ or adolescent/ or exp pediatrics/ or child, abandoned/ or exp child, exceptional/ or child, orphaned/ or child, unwanted/ or minors/
16. (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or pre-term or preterm* or premature birth* or NICU or preschool* or pre-school* or kindergarten* or kindergarden* or elementary school* or nursery school* or (day care* not adult*) or schoolchild* or toddler* or boy or boys or girl* or middle school* or pubescen* or juvenile* or teen* or youth* or high school* or adolesc* or pre-pubesc* or prepubesc*).mp. or (child* or adolesc* or pediat* or paediat*).jn.
17. 15 or 16
18. 11 and 14 and 17

CINAHL (EBSCO) search strategy

1. (MH "deglutition disorders") OR TX (dysphagia* OR deglutition OR swallow OR swallows OR swallowing OR feeding* OR odynophagia* OR feeding* OR regurgit*) OR TX (deglut* N3 (abnormal* OR disorder* OR dysfunc* OR impair*)) OR TX (oropharynx* OR trachea* OR lung* OR pulmon*) N3 aspirat*)) OR TX (pharyng* N3 (dysphag* OR dysfunct* OR disorder* OR impair*))
2. ((MH "oropharynx") OR TX (oropharyn*)) AND TX (dysphag* OR disorder* OR dysfunc* OR impair*)
3. 1 or 2
4. (MH "electric stimulation therapy")
5. TX (electrostimulation or NMES or "electrical stimulation" or "electric stimulation" or electrotherapy or "electro therapy")
6. 4 or 5
7. (MH "Child+") OR (MH "Child Behavior+") OR (MH "Child Abuse, Sexual") OR (MH "Child Behavior Disorders+") OR (MH "Child, Medically Fragile") OR (MH "Child Day Care") OR (MH "Child Passenger Safety") OR (MH "Child Abuse+") OR (MH "Child Custody") OR (MH "Child Behavior Checklist") OR (MH "Child Nutritional Physiology+")
8. (MH "Child Development Disorders, Pervasive+") or (MH "Adolescence+") or (MH "Minors (Legal)") or (MH "Child Mortality")
9. SO (child* or pediatric* or paediatric* or adolescent)
10. TX (pediatric* or paediatric* or child* or newborn* or congenital* or infan* or baby or babies or neonat* or "pre-term" or preterm or "premature birth*" or NICU or preschool* or "preschool*"

or kindergarten* or "elementary school*" or "nursery school*" or schoolchild* or toddler* or boy or boys or girl* or "middle school*" or pubescen* or juvenile* or teen* or youth* or "high school*" or adolesc* or prepubesc* or "pre-pubesc*")

11. or/7-10
12. 3 and 6 and 11

Scopus search strategy

((TITLE-ABS-KEY (dysphagia* OR deglutition OR swallow OR swallows OR swallowing OR feeding* OR odynophagia* OR feeding* OR regurgit*)) OR (TITLE-ABS-KEY (deglut* W/3 (abnormal* OR disorder* OR dysfunc* OR impair*))) OR (TITLE-ABS-KEY ((oropharynx* OR trachea* OR lung* OR pulmon*) W/3 aspirat*)) OR (TITLE-ABS-KEY (pharyng* W/3 (dysphag* OR dysfunct* OR disorder* OR impair*))) OR (TITLE-ABS-KEY (oropharyn*) AND TITLE-ABS-KEY (dysphag* OR disorder* OR dysfunc* OR impair*))) AND ((TITLE-ABS-KEY (electrostimulation OR nmes OR "electrical stimulation" OR "electric stimulation" OR electrotherapy OR "electro therapy")) AND (TITLE-ABS-KEY (pediatric* OR paediatric* OR child* OR newborn* OR congenital* OR infan* OR baby OR babies OR neonat* OR "pre term" OR preterm* OR "premature birth" OR nicu OR preschool* OR "pre school*" OR kindergarten* OR "elementary school*" OR "nursery school*" OR schoolchild* OR toddler* OR boy OR boys OR girl* OR "middle school*" OR pubescen* OR juvenile* OR teen* OR youth* OR "high school*" OR adolesc* OR prepubesc* OR "pre pubesc*")))

Appendix B. Citation information for unpublished studies and abstracts

Author	Year	Title	Journal
Biernacka	2019	Evaluation of physical and emotional well-being of children undergoing electrostimulation of swallowing	J Pediatr Gastroenterol Nutr
Winnicka	2019	Improving swallowing function due to electrical stimulation in group of children-preliminary results from an innovative method of treatment	J Pediatr Gastroenterol Nutr
Winnicka	2019	Improvement of oral feeding due to electrical stimulation in children with primary dysphagia	J Pediatr Gastroenterol Nutr
Radika	2018	Role of NMES in dysphagia management for children with hypertonia and dystonia: a case study	Dysphagia
Biernacka	2017	Evaluation of physical and emotional comfort of children undergoing electrostimulation of swallowing- Initial report	J Pediatr Gastroenterol Nutr
Andreoli	2016	Neuromuscular stimulation and feeding in medically complex children	Otolaryngol Head and Neck Surg
Demir	2015	Swallowing function of a pierre robin sequence with absence epiglottis: A case report	Dysphagia
Mitchell	2010	The effectiveness of NMES (VitalStim) therapy in the neonatal population	Unpublished
Christiaanse	2005	Experience with pharyngeal electrical stimulation: A new treatment modality for the management of pediatric dysphagia	Dev Med Child Neurol
Chatburn	2000	Electrical stimulation for treatment of dysphagia in children failing conventional therapy	Respiratory Care

Appendix C. Study Funding Source

Study	Year	Funding Source
Andreoli[35]	2019	Not indicated
Christiaanse[36]	2011	No financial support
El-Sheikh[31]	2020	Not indicated
Gao[28]	2018	Not indicated
Lv[27]	2019	National Natural Science Foundation of China Early Career Fund, Young Talents of Science, Education and Health in Jiangsu Province
Ma[32]	2019	Not indicated
Marcus[33]	2019	Peer-reviewed grant from the Department of Paediatrics, The Hospital for Sick Children, Toronto
Rice[34]	2012	Not indicated
Serel Arslan[29]	2017	No financial support
Song[30]	2015	No financial support